

Data Science

Assignment – 3



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1 Knowledge discovery

1. Explain in your own words what Fayyad et al. (1996) mean when they refer to the process of “identifying valid, novel, potentially useful, and ultimately understandable patterns in data”. Provide examples for each adjective used in their definition.

Answer: “Knowledge Discovery in Databases (KDD) is the non-trivial process of identifying valid, novel, potentially useful, and ultimately understandable patterns in data.”

- Here data is a set of facts and pattern is an expression that repeats itself as per given set of rules or conditions.
- Discovered patterns must be **valid** and reliable on new data. Pattern should hold in general. Eg., Patterns with missing values or random noises are not preferred.
- Patterns expected to be **novel** to system and the use. Pattern should be unknown beforehand. Uniqueness of patterns leads us to the more efficiency process and new discovery.
- **Potentially useful** patterns are those, that can lead to some benefit to the user/task. Actions can be derived from patterns.
- Here **ultimately understandable** can be said as simple patterns that are very easy to understand (number of bits to describe pattern), If it not immediately, then after some processing. We can interpret and comprehend the patterns.

So, basically, KDD is process of identifying the patterns that are valid in general and previously unknown as well as potentially useful and easy to understand.

2 Data Mining Tasks

Q. Explain the differences between supervised and unsupervised learning. Write approx. 400 words.

Solution:

Supervised Learning:

One of the subcategories of machine learning is supervised learning. It produces output by training algorithms on a labeled dataset. Algorithms can be trained on these datasets to accurately classify data and predict the outcome.

Supervised learning has a feedback mechanism to check the accuracy of output and It is mainly used for prediction.

The training of the data is the purpose of supervised learning. So that it can accurately predict the outcome of future datasets. Different algorithms are included in supervised learning. e.g, SVM, Bayesian logic, Linear Regression, etc.

Classification and regression are two supervised machine learning techniques. The classification algorithm takes the input value and classifies the dataset into different classes/ categories. i.e. separating apples and bananas. The real world example can be identifying spam mails and important mails in our inbox.

Whereas, the output values can be predicted using regression. Additionally, the predicted output values are represented by actual numbers. Regression is used to predict the price of stocks, real estate, and some products in the real world.

Image classification, weather forecasting, price prediction, text categorization, spam detection, and signature recognition are a few examples of supervised machine learning applications.

Unsupervised Learning:

Unsupervised learning is another subcategory of machine learning. In contrast to supervised learning, unsupervised learning does not use labeled data. The models of unsupervised learning algorithms are trained on unlabeled data. An unsupervised learning algorithm uses similarities and hidden patterns in the input data to try to predict the output.

Unsupervised learning is mostly used for analysis and lacks a feedback mechanism to verify output accuracy.

Numerous algorithms are included in unsupervised learning. Examples include the KNN algorithm, the Apriori algorithm, and Clustering.

Clustering, Association and Dimensionality reduction are three unsupervised machine learning techniques. Using unlabeled data, clustering is an unsupervised learning method that groups the data into distinct clusters based on similarities or hidden patterns. For instance, it can be used to locate a group of pictures of fruits that are similar.

Association, it is used to establish a connection between various data objects from an input dataset. We can benefit from these relationships. For instance, the majority of e-commerce websites use customer habits to enhance their marketing strategies.

Dimensionality reduction, on the other hand, uses feature extraction from a specific input dataset to reduce the dimensionality of the data for improved comprehension.

Applications of unsupervised machine learning include anomaly detection, customer behaviour prediction, and recommendation engines.

3 Simple Linear Regression with R



```
title: "Team: Fowler, Question:3"
output: html_notebook
---
```

```
```{r}
#Exercise:- Have a look at the â€˜faithfulâ€™ dataset (use the help
?faithful to find out more about the dataset). Use the lm() function to
build a simple regression model to predict the eruption time based on the
waiting time to the next eruption. What is the predicted eruption time given
a waiting time of 80 mins?
library (tidyverse)
data ("faithful")
str (faithful)
```

```{r}
summary (faithful)
```

```{r}
#The scatterplot
ggplot (data = faithful , aes(x = waiting , y = eruptions)) + geom_point ()
```

```{r}
#The lm() function to build a simple regression model to predict the
eruption time based on the waiting time to the next eruption
faithfulModel = lm(eruptions~waiting, data=faithful)
summary (faithfulModel)
```

```{r}
#The model
ggplot (data = faithful , aes(x = waiting , y = eruptions
))+geom_point()+geom_abline (mapping=aes (slope =
faithfulModel$coefficients[2],intercept = faithfulModel$coefficients[1]),
color='red')
```

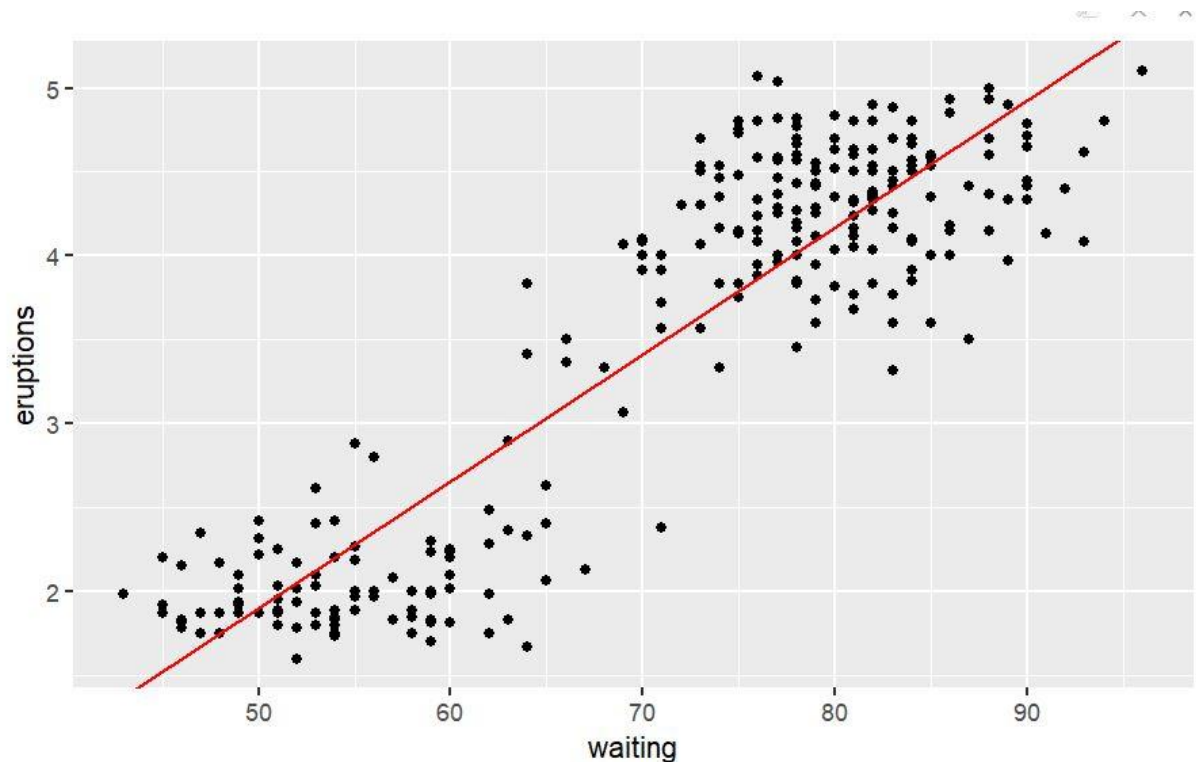
```{r}
#Residual values
plot (faithfulModel , which =1)
```

```{r}
#The diagnostic
layout (matrix (1:6 , ncol =2, byrow = TRUE))
plot (faithfulModel , 1:6)
```

```{r}
#The predicted eruption time given a waiting time of 80 mins is 4.17622
eurption <-data.frame (waiting = 80)
predict (faithfulModel , newdata = eurption)
```

```{r}
The avg eruption time lies between 4.104848 and 4.247592 if waiting time
is 80 mins
predict (faithfulModel , newdata = eurption , interval ="confidence")
```

```{r}
The eruption time lies between 3.196089 and 5.156351 for 95% of the cases
where the waiting time is 80 mins
predict (faithfulModel , newdata = eurption , interval ="prediction")
```
```



#The predicted eruption time given a waiting time of 80 mins is 4.17622

```
eurption <- data.frame ( waiting = 80)
predict ( faithfulModel , newdata = eurption )
```

```
1
4.17622
```

```
{r}
# The avg eruption time lies between 4.104848 and 4.247592 if waiting time
is 80 mins
predict ( faithfulModel , newdata = eurption , interval = "confidence")
```

```
      fit      lwr      upr
1 4.17622 4.104848 4.247592
```

```
{r}
# The eruption time lies between 3.196089 and 5.156351 for 95% of the cases
where the waiting time is 80 mins
predict ( faithfulModel , newdata = eurption , interval = "prediction")
```

```
      fit      lwr      upr
1 4.17622 3.196089 5.156351
```

4 Multiple Linear Regression using R



```
---
title: "Team: Fowler, Question:4"
output: html_notebook
```

```
```{r}
#Exercise: Build a simple linear regression with just the wt variable â€œ
how does the quality and fit of the model compare with using all variables
and the selected subset using step()? What about other subsets?

data ("mtcars")

#Model 1 with just -> wt
mpgModelWt <-lm(formula =mpg~wt,
data = mtcars)
summary (mpgModelWt)
plot (mpgModelWt , which =1)
```

```{r}
#Model 2, Model with all the variables
mpgModel <-lm(formula =mpg~cyl+disp+hp+drat+wt+qsec+vs+am+gear+carb ,
data = mtcars)
summary (mpgModel)
plot (mpgModel , which =1)
```

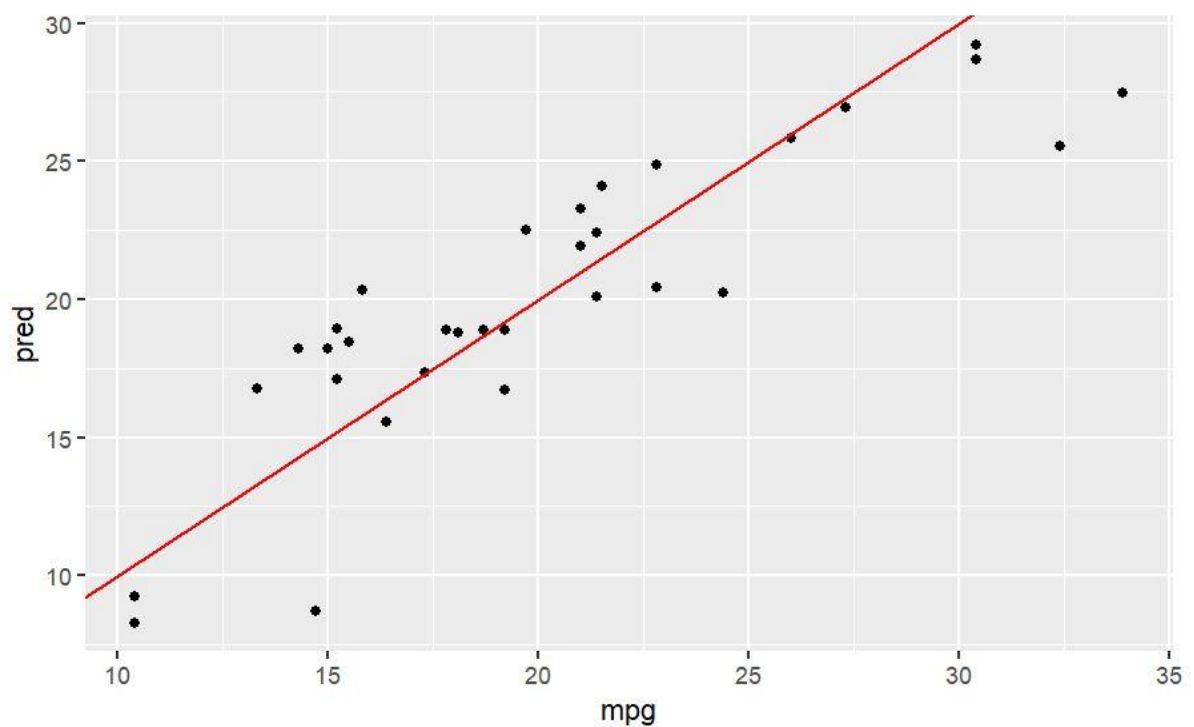
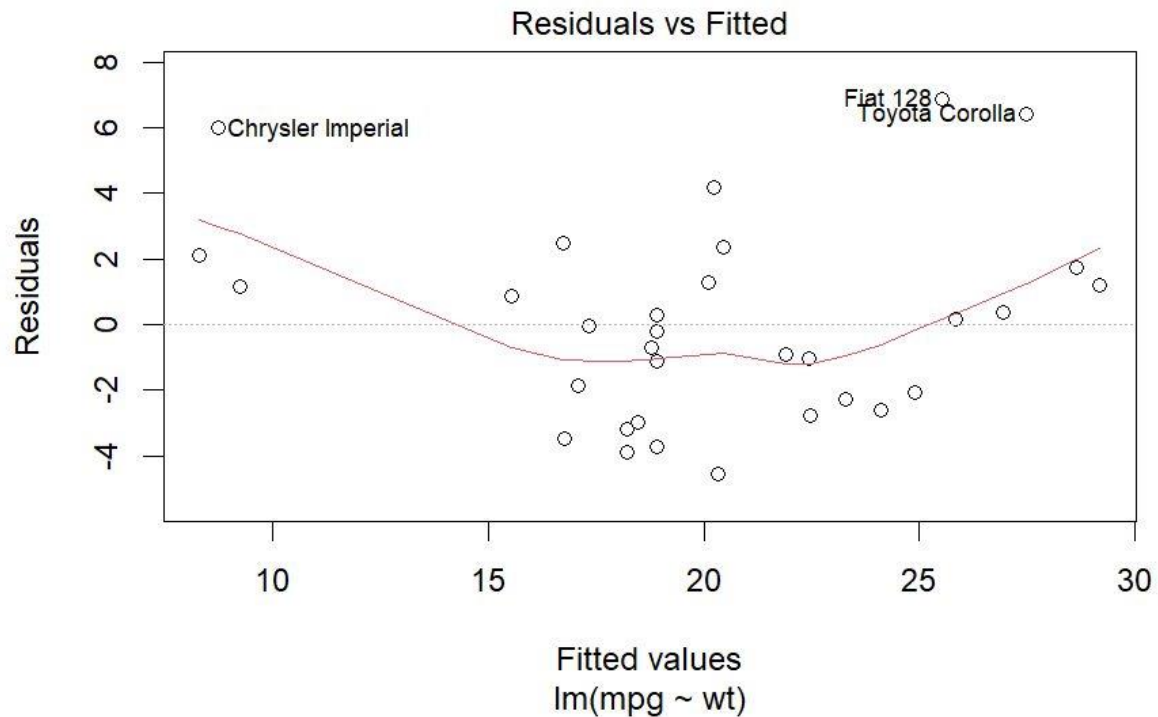
```{r}
#Model 3, Model with subset
reducedModel <-step (mpgModel , direction ="backward")
summary (reducedModel)
plot (mpgModel , which =1)
```

```{r}
#The quality and fit of the model with just wt
mpgPredictions <-data.frame (mpg= mtcars$mpg , pred =
mpgModelWt$fitted.values)
mpgPredictions
ggplot (data = mpgPredictions , aes(x = mpg , y = pred)) + geom_point() +
geom_abline (mapping =aes (slope =1, intercept =0) , color ='red ')
```

```{r}
#The quality and fit of the model with all
mpgPredictions1 <-data.frame (mpg= mtcars$mpg , pred =
mpgModel$fitted.values)
mpgPredictions1
ggplot (data = mpgPredictions1 , aes(x = mpg , y = pred)) + geom_point() +
geom_abline (mapping =aes (slope =1, intercept =0) , color ='red ')
```

```{r}
#The quality and fit of the model with subset
mpgPredictions2 <-data.frame (mpg= mtcars$mpg , pred =
reducedModel$fitted.values)
mpgPredictions2
ggplot (data = mpgPredictions2 , aes(x = mpg , y = pred)) + geom_point() +
geom_abline (mapping =aes (slope =1, intercept =0) , color ='red ')
```

```{r}
Answer:
After comparing the above three models, by examining the change in R2 and
adjusted R2 values of all three regression equations we can conclude that
the equation one, that is $\text{mpg} \sim \text{wt}$ is the best one, because the
percentage change in R2 and adjusted R2 values is least for this
regression equation.
```
```



```
##{r}
# Answer:
# After comparing the above three models, by examining the change in R2 and
adjusted R2 values of all three regression equations we can conclude that
the equation one, that is "mpg~wt" is the best one, because the percentage
change in R2 and adjusted R2 values is least for this regression equation.
```